Triangle Books with a Two-Edge Tail: A Sharp Operator-Square Inequality

Abstract

Let R_m be a book of $m \geq 1$ triangles on one common spine edge, with one otherwise disjoint path of length two attached to a spine endpoint. We prove the chromatic-polynomial graphon bound for every R_m and every $p \geq 1/2$. The new input is a sharp inequality for the weight $A = T_W d$:

$$\int A\tau \geq (2p-1) \int Ad.$$

Unlike a monotonicity assertion for the weight A, this follows from an exact Hilbert-space square after writing $d = p\mathbf{1} + g$. Moment convexity under the positive measure $W(x,y)A(x)\,\mathrm{d}\mu(x)\mathrm{d}\mu(y)$ then treats all pages at once. The case $m = 2$ proves the formerly open NetworkX Atlas graph 120.

1 The family and its exact target

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be a symmetric measurable graphon. Write

$$T = T_W, \qquad p = \int_{\Omega^2} W\,\mathrm{d}\mu^2, \qquad d = T\mathbf{1}, \qquad A = Td = T^2\mathbf{1}.$$

We also use the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z)$$

and the common-neighbour kernel

$$S(x,y) = \int_{\Omega} W(x,z)W(y,z)\,\mathrm{d}\mu(z).$$

All these functions are measurable and take values in $[0,1]$.

For $m \geq 1$, define R_m on a spine ab, page vertices $z_1, \ldots, z_m$, and tail vertices u, v. Its edges are

$$ab, \quad az_i, bz_i \ (1 \leq i \leq m), \quad au, uv.$$

Thus R_m is a book of m triangles with one two-edge tail at the spine endpoint a. A proper colouring first colours the ordered spine, then each page independently, and finally the two tail vertices. Hence

$$\chi_{R_m}(x) = x(x-1)^3(x-2)^m, \qquad \chi(R_m) = 3. \tag{1}$$

Putting $q = 1 - p$ and substituting $x = 1/q$ gives

$$\Phi_{R_m}(p) = q^{m+4}\chi_{R_m}(1/q) = p^3(2p-1)^m. \tag{2}$$

The formula is polynomial and therefore also gives the value at $p = 1$. The required interval is $p \in [1/2, 1]$.

2 A sharp inequality for the two-edge-tail weight

The next lemma is the only new analytic ingredient. It is important that its proof does not insert A into the previously established degree-weighted rooted-Goodman lemma: A is not a pointwise monotone function of the complement degree, so that substitution would be invalid.

Lemma 2.1 ($T_W d$-weighted rooted triangle)**.** *If $p \geq 1/6$, then*

$$\int_\Omega A(x)\tau(x)\,\mathrm{d}\mu(x) \geq (2p-1)\int_\Omega A(x)d(x)\,\mathrm{d}\mu(x). \tag{3}$$

Proof. For $a, b \in [0,1]$ one has $ab \geq a+b-1$. Apply this to $W(x,y), W(x,z)$, multiply by the nonnegative $W(y,z)$, and integrate in y, z. Symmetry and Fubini give the pointwise estimate

$$\tau(x) \geq 2A(x) - p. \tag{4}$$

Since $A \geq 0$, it follows that

$$\int A\tau \geq 2\|A\|_2^2 - p\|d\|_2^2, \tag{5}$$

where we used

$$\int A = \langle Td, \mathbf{1}\rangle = \langle d, T\mathbf{1}\rangle = \|d\|_2^2.$$

It remains to prove

$$E := 2\|A\|_2^2 - p\|d\|_2^2 - (2p-1)\langle A, d\rangle \geq 0. \tag{6}$$

Put $g = d - p\mathbf{1}$ and $h = Tg$. Then $\int g = 0$ and

$$A = T(p\mathbf{1} + g) = pd + h.$$

The coefficient of $\|d\|_2^2$ cancels exactly in (6), leaving

$$E = 2\|h\|_2^2 + (2p+1)\langle h, d\rangle. \tag{7}$$

Self-adjointness of T and $T\mathbf{1} = d$ show that

$$\langle h, \mathbf{1}\rangle = \langle Tg, \mathbf{1}\rangle = \langle g, d\rangle = \|g\|_2^2.$$

Consequently

$$\langle h, d\rangle = p\|g\|_2^2 + \langle h, g\rangle.$$

Substitution in (7) and completion of the square give the exact identity

$$E = 2\left\|h + \frac{2p+1}{4}g\right\|_2^2 + \frac{(2p+1)(6p-1)}{8}\|g\|_2^2. \tag{8}$$

Both coefficients are nonnegative for $p \geq 1/6$. Combining (5)–(8) proves (3). □

Remark 2.2 (Why the square is substantive)**.** *No positivity of the graphon operator is assumed; a nonnegative symmetric kernel can have negative eigenvalues. Identity* (8) *controls those spectral directions through a quadratic polynomial with negative discriminant when $p > 1/6$. Thus the proof is valid for arbitrary graphons, not only positive-semidefinite or regular kernels.*

3 The book-tail theorem

Define

$$B = \int_\Omega A(x)d(x)\,\mathrm{d}\mu(x), \qquad F = \int_\Omega A(x)\tau(x)\,\mathrm{d}\mu(x).$$

The quantity $B = t(P_4, W)$, while $F = t(R_1, W)$ is the density of a triangle with one two-edge tail.

Lemma 3.1 (Sharp lower bound for the first page)**.** *For $p \in [1/2, 1]$,*

$$F \geq p^3(2p-1).$$

Proof. Let $M = \int A = \int d^2$. From (4), Cauchy–Schwarz, and Jensen,

$$F \geq 2\int A^2 - p\int A \geq 2M^2 - pM.$$

Also $M \geq p^2$. The function $r \mapsto 2r^2 - pr$ is increasing for $r \geq p^2$ when $p \geq 1/2$, because there $4r - p \geq 4p^2 - p \geq 0$. Hence

$$F \geq 2p^4 - p^3 = p^3(2p-1).$$

□

Theorem 3.2 (Triangle-book two-edge-tail theorem)**.** *Let $m \geq 1$. Every graphon W of edge density $p \in [1/2, 1]$ satisfies*

$$\boxed{t(R_m, W) \geq p^3(2p-1)^m = \Phi_{R_m}(p).}$$

Proof. Because $p > 0$, the forest inequality gives $B = t(P_4, W) \geq p^3 > 0$. Define a finite positive measure on Ω^2 by

$$\mathrm{d}\nu(x, y) = W(x, y)A(x)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y).$$

Its total mass is

$$\nu(\Omega^2) = \int W(x, y)A(x)\,\mathrm{d}\mu^2 = B.$$

Integrating all page vertices and the two tail vertices in the defining homomorphism integral gives the exact identity

$$t(R_m, W) = \int_{\Omega^2} S(x, y)^m\,\mathrm{d}\nu(x, y). \tag{9}$$

Moreover, symmetry and Fubini give

$$\int_{\Omega^2} S(x, y)\,\mathrm{d}\nu(x, y) = \int_\Omega A(x)\tau(x)\,\mathrm{d}\mu(x) = F. \tag{10}$$

Apply Jensen's inequality to the convex function $s \mapsto s^m$ under the probability measure ν/B. Equations (9)–(10) yield

$$t(R_m, W) \geq B\left(\frac{F}{B}\right)^m = F\left(\frac{F}{B}\right)^{m-1}. \tag{11}$$

Lemma 2.1 gives $F/B \geq 2p-1$, and Lemma 3.1 gives $F \geq p^3(2p-1)$. Since $2p-1 \geq 0$, substitution in (11) proves

$$t(R_m, W) \geq p^3(2p-1)^m.$$

This is the target in (2). □

At the threshold $p = 1/2$ the target is zero; the proof above remains valid and uses no division by $2p-1$. At $p = 1$, $W = 1$ almost everywhere and both sides equal one. For every integer $k \geq 2$, the balanced complete k-partite graphon has $p = 1 - 1/k$ and realizes equality: along a graphon edge, $S(x, y) = 1 - 2/k = 2p-1$, while $d = p$ and $A = p^2$ almost everywhere.

4 Atlas 120 and formalization boundary

Corollary 4.1 (Atlas 120)**.** *NetworkX Atlas graph* 120*, with graph6 code* `EB}?`*, is positive. It has order six, size seven, chromatic number three, and*

$$\chi_{F_{120}}(x) = x(x-1)^3(x-2)^2.$$

Every graphon of edge density $p \in [1/2, 1]$ *satisfies*

$$t(F_{120}, W) \geq p^3(2p-1)^2 = \Phi_{F_{120}}(p).$$

Proof. Its NetworkX edge list is

$$04,\ 05,\ 13,\ 14,\ 23,\ 24,\ 34.$$

Take spine $ab = 43$, page vertices $1, 2$, and tail $4 - 0 - 5$. This is exactly R_2, so Theorem 3.2 applies. □

Every use of Tonelli–Fubini above concerns bounded nonnegative measurable integrands. The operator calculation uses only bounded self-adjointness of T_W on L^2 and never assumes that T_W is positive semidefinite. The sole previously accepted density input is the forest bound for P_4, used only to certify that the normalizing mass B is positive; its full sharp lower value is not substituted into the final estimate.

For formal verification, the reusable new lemma is Lemma 2.1, including the pointwise triangle reduction and the exact operator-square identity (8). The remaining steps are the two Fubini identities, probability-measure normalization, Jensen for integer moments, and the scalar first-page estimate. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph120.lean`.

The family requires every page to use the same spine edge and the two-edge tail to start at a spine endpoint. It does not cover a tail at a page vertex (in particular Atlas 123), two-edge tails on an arbitrary chordal core, or mixed maximal-clique sizes in general.

The graph, chromatic, target, Hilbert-square, moment, equality, and Atlas identities were reconstructed in exact rational arithmetic.

5 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/BookTail/{Core,Rows}.lean`, ending at `satisfiesLowerBound_120`. It follows this note step for step except at one place, and the difference is worth stating precisely so that neither document is read as a paraphrase of the other.

The departure: Lemma 2.1 on $[1/2, 1]$ **only.** This note proves

$$\int A\tau \geq (2p-1) \int Ad \qquad \text{for every } p \geq 1/6,$$

through the exact operator square (8). The formalization proves the same inequality *only for* $p \geq 1/2$, and by a different argument. Write

$$M = \int A = \|d\|_2^2, \qquad N = \|A\|_2^2, \qquad B = \langle A, d \rangle .$$

Three applications of Cauchy–Schwarz give $B^2 \le NM$, $N \ge M^2$ and $M \ge p^2$, and then the polynomial identity

$$(2N - pM)^2 - (2p-1)^2 NM = (4N - M)(N - p^2 M) \tag{12}$$

has both right-hand factors nonnegative: $N \ge M^2 \ge p^2 M$ gives the second, and $M \ge p^2 \ge 1/4$ gives $4N \ge 4M^2 \ge M$ for the first. With $2N - pM \ge 0$ and $2p - 1 \ge 0$ this yields $2N - pM \ge (2p-1)B$, which combined with (5) is (3).

Both routes are correct; they differ in reach and in cost. The operator square is the stronger statement, valid on $[1/6, 1]$, and it is the one to formalize if that range is ever wanted — it needs the six-term expansion of $\int (A + ud + v)^2$ and the substitutions $\langle h, \mathbf{1}\rangle = \|g\|_2^2$, $A = pd + h$. The factorization (12) needs no operator language at all, and $p \ge 1/2$ is exactly the interval on which the catalogue proposition is asserted for a graph of chromatic number three. No Atlas row — present or possible, since $\chi(R_m) = 3$ for every m — requires the extra range, so the formalization takes the cheaper route.

Scope: only $m \le 2$ is formalized. Theorem 3.2 is proved here for every $m \ge 1$, by Jensen for $s \mapsto s^m$ under ν/B. The Lean file proves the case $m = 2$, where that Jensen step is a single Cauchy–Schwarz, and does not state the general m. This loses nothing for the catalogue: R_1 is the five-vertex triangle with a two-edge tail, already `verified` through the rooted-tree route, and R_3 already has seven vertices.

Everything else transcribes directly. The pointwise estimate (4) is `Link.rootedTriangle_ge`, which predates this note; $\int A = \int d^2$ is `BookTail.integral_pathOp`, one application of self-adjointness; (9) and (10) are `BookTail.integral_tailWeight` and `integral_tailWeight_mul_pageOp`; Lemma 3.1 is `BookTail.firstPage_bound`, with the monotonicity of $r \mapsto 2r^2 - pr$ replaced by the equivalent factorization $2M^2 - pM - p^3(2p-1) = (M - p^2)\big(2(M + p^2) - p\big)$. The final division by B is done as `le_of_mul_le_mul_right`, so the argument never divides by $2p - 1$, which vanishes at the threshold.